

qBitTorrent Backend Migration: Python/FastAPI to Go/Gin

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1. Project Analysis & Pre-Migration Checklist

1.1 Source of Truth

- **OpenAPI Specification:** `http://localhost:7187/openapi.json`
This file defines every endpoint, request/response schema, and error code.
Action: Export it and keep it open for reference.
- **Existing Shell Scripts:** `start.sh`, `install-plugin.sh`, `stop.sh` remain unchanged.
- **Angular Frontend:** Located in `frontend/`. No code changes required.

1.2 Python Code Audit (Hypothetical to Based on OpenAPI)

The Python backend consists of:

- `main.py` to FastAPI application entry point.
- `app/api/search.py` to `/api/v1/search` endpoint with SSE streaming.
- `app/api/hooks.py` to Hook registration and trigger logic.
- `app/api/bridge.py` to Private tracker authentication and download.
- `app/core/config.py` to Environment variable loading.
- `app/models/*.py` to Pydantic models for requests/responses.

1.3 Dependencies to Replace

Python Dependency Go Replacement

fastapi	gin-gonic/gin
uvicorn	Built-in net/http server
pydantic	Struct tags + validator package
httpx/requests	net/http + custom client
asyncio	Goroutines + channels
python-dotenv	godotenv or viper
sse-starlette	Manual SSE with http.Flusher

1.4 Plugin System Status

- **Nova3 search plugins** are Python scripts executed **by the qBittorrent client itself**, not by our backend.
 - Our backend only calls qBittorrent's Web API (/api/v2/search/start, /api/v2/search/results, etc.).
 - **Conclusion:** Plugins will work unchanged. The migration only affects the orchestrator.
-

2. Go Project Initialization & Structure

2.1 Initialize Module

```
mkdir qBitTorrent-go && cd qBitTorrent-go
go mod init github.com/milos85vasic/qBitTorrent-go
```

2.2 Directory Layout

```
.
├── cmd/
│   ├── qbittorrent-proxy/          # Main REST API server
│   ├── main.go
│   └── webui-bridge/              # Private tracker bridge (separate
│       └── main.go
├── internal/
│   ├── api/                      # Gin handlers grouped by feature
│   ├── search.go
│   ├── hooks.go
│   ├── bridge.go
│   ├── health.go
│   ├── client/                  # qBittorrent Web API client
│   ├── client.go
│   ├── auth.go
│   ├── torrents.go
│   ├── search.go
│   ├── config/                  # Configuration structs and loader
│   ├── config.go
│   ├── models/                  # Shared data structures
│   └── torrent.go
```

```

â",    â",    â"œâ"€â"€ search.go
â",    â",    â""â"€â"€ hook.go
â",    â"œâ"€â"€ service/                # Business logic
â",    â",    â"œâ"€â"€ merge_search.go
â",    â",    â"œâ"€â"€ sse_broker.go
â",    â",    â""â"€â"€ bridge_auth.go
â",    â""â"€â"€ middleware/              # Gin middleware
â",        â"œâ"€â"€ cors.go
â",        â"œâ"€â"€ logging.go
â",        â""â"€â"€ auth.go
â"œâ"€â"€ pkg/                            # Public reusable packages (if any)
â"œâ"€â"€ scripts/                        # Build/deploy scripts
â",    â""â"€â"€ build.sh
â"œâ"€â"€ frontend/                        # Angular frontend (unchanged)
â"œâ"€â"€ .env.example
â"œâ"€â"€ Dockerfile
â"œâ"€â"€ docker-compose.yml
â"œâ"€â"€ go.mod
â""â"€â"€ go.sum

```

3. Data Models Translation (Pydantic â†’ Go Structs)

3.1 Example: Torrent Model

Python (Pydantic) â€“ app/models/torrent.py

```

from pydantic import BaseModel
from typing import Optional

class Torrent(BaseModel):
    hash: str
    name: str
    size: int
    progress: float
    state: Optional[str] = None
    category: Optional[str] = None

```

Go â€“ internal/models/torrent.go

package models

```

type Torrent struct {
    Hash      string `json:"hash"`
    Name      string `json:"name"`
    Size      int64  `json:"size"`
    Progress  float64 `json:"progress"`
    State     *string `json:"state,omitempty"`
    Category  *string `json:"category,omitempty"`
}

```

3.2 Search Request/Response Models

From OpenAPI: - POST /api/v1/search expects: json { "query": "ubuntu", "plugins": ["all"], "category": "all" } - Response: Stream of SSE events with Torrent objects.

Go â€“ internal/models/search.go

```
package models

type SearchRequest struct {
    Query      string    `json:"query" binding:"required"`
    Plugins    []string  `json:"plugins" binding:"required"`
    Category   string    `json:"category"`
    Limit      int       `json:"limit,omitempty"`
}

type SearchResult struct {
    Tracker     string    `json:"tracker"`
    Torrents    []Torrent `json:"torrents"`
}
```

3.3 Hook Model

Go â€“ internal/models/hook.go

```
package models

type Hook struct {
    ID          string    `json:"id"`
    URL         string    `json:"url" binding:"required,url"`
    Secret      string    `json:"secret,omitempty"`
    Events      []string  `json:"events" binding:"required"`
    Headers     map[string]string `json:"headers,omitempty"`
    Enabled     bool      `json:"enabled"`
}
```

4. Configuration Management

4.1 Configuration Struct

File: internal/config/config.go

```
package config

import (
    "log"
    "os"
    "strconv"

    "github.com/joho/godotenv"
)
```

```

type Config struct {
    QBittorrentURL      string
    QBittorrentUsername string
    QBittorrentPassword string
    ServerPort          string
    BridgePort          string
    LogLevel            string
    SSETimeout          int
    PluginTimeout       int
    PrivateTrackerConfig map[string]TrackerAuth
}

type TrackerAuth struct {
    Username string
    Password string
    LoginURL string
}

func Load() *Config {
    _ = godotenv.Load() // Ignore error if .env missing

    return &Config{
        QBittorrentURL:      getEnv("QBITTORRENT_URL", "http://localhost:8
        QBittorrentUsername: getEnv("QBITTORRENT_USERNAME", "admin"),
        QBittorrentPassword:  getEnv("QBITTORRENT_PASSWORD", "adminadmin"),
        ServerPort:          getEnv("SERVER_PORT", "7187"),
        BridgePort:          getEnv("BRIDGE_PORT", "7188"),
        LogLevel:            getEnv("LOG_LEVEL", "info"),
        SSETimeout:          getEnvAsInt("SSE_TIMEOUT", 30),
        PluginTimeout:       getEnvAsInt("PLUGIN_TIMEOUT", 10),
        // Load private tracker credentials from env like TRACKER_IPT_USER
        PrivateTrackerConfig: loadTrackerAuth(),
    }
}

func getEnv(key, fallback string) string {
    if value, exists := os.LookupEnv(key); exists {
        return value
    }
    return fallback
}

func getEnvAsInt(key string, fallback int) int {
    if value, exists := os.LookupEnv(key); exists {
        if i, err := strconv.Atoi(value); err == nil {
            return i
        }
    }
    return fallback
}

func loadTrackerAuth() map[string]TrackerAuth {
    // Implementation to read env vars like TRACKER_IPT_USERNAME etc.

```

```

    // Returns map[trackerID]TrackerAuth
    return make(map[string]TrackerAuth)
}

```

4.2 Environment File Example

.env.example

```

QBITTORRENT_URL=http://localhost:8080
QBITTORRENT_USERNAME=admin
QBITTORRENT_PASSWORD=adminadmin
SERVER_PORT=7187
BRIDGE_PORT=7188
LOG_LEVEL=debug
SSE_TIMEOUT=30
PLUGIN_TIMEOUT=10
TRACKER_IPT_USERNAME=your_ipt_user
TRACKER_IPT_PASSWORD=your_ipt_pass

```

5. qBittorrent API Client Implementation

5.1 Client Structure

File: internal/client/client.go

```

package client

import (
    "bytes"
    "encoding/json"
    "fmt"
    "io"
    "net/http"
    "net/http/cookiejar"
    "net/url"
    "sync"
    "time"
)

type Client struct {
    BaseURL      *url.URL
    HTTPClient   *http.Client
    mu           sync.RWMutex
    sid          string // qBittorrent session cookie value
}

func NewClient(baseURL, username, password string) (*Client, error) {
    u, err := url.Parse(baseURL)
    if err != nil {
        return nil, err
    }
}

```

```

    jar, _ := cookiejar.New(nil)
    c := &Client{
        BaseURL: u,
        HTTPClient: &http.Client{
            Jar: jar,
            Timeout: 30 * time.Second,
        },
    }

    if err := c.Login(username, password); err != nil {
        return nil, err
    }
    return c, nil
}

```

5.2 Authentication

File: internal/client/auth.go

```

package client

import (
    "fmt"
    "net/url"
)

func (c *Client) Login(username, password string) error {
    loginURL := c.BaseURL.ResolveReference(&url.URL{Path: "/api/v2/auth/login"})
    data := url.Values{}
    data.Set("username", username)
    data.Set("password", password)

    resp, err := c.HTTPClient.PostForm(loginURL.String(), data)
    if err != nil {
        return err
    }
    defer resp.Body.Close()

    if resp.StatusCode != http.StatusOK {
        return fmt.Errorf("login failed: %s", resp.Status)
    }

    // Extract SID cookie (qBittorrent sets "SID")
    for _, cookie := range c.HTTPClient.Jar.Cookies(loginURL) {
        if cookie.Name == "SID" {
            c.mu.Lock()
            c.sid = cookie.Value
            c.mu.Unlock()
            break
        }
    }
    return nil
}

```

```
func (c *Client) IsAuthenticated() bool {
    c.mu.RLock()
    defer c.mu.RUnlock()
    return c.sid != ""
}
```

5.3 Search Endpoints

File: internal/client/search.go

```
package client
```

```
import (
    "encoding/json"
    "fmt"
    "net/http"
    "net/url"
    "strconv"

    "github.com/milos85vasic/qBitTorrent-go/internal/models"
)
```

// StartSearch starts a new search job and returns its ID.

```
func (c *Client) StartSearch(query string, plugins []string, category string) (int, error) {
    searchURL := c.BaseURL.ResolveReference(&url.URL{Path: "/api/v2/search"})
    data := url.Values{}
    data.Set("pattern", query)
    data.Set("category", category)
    for _, p := range plugins {
        data.Add("plugins", p)
    }

    resp, err := c.HTTPClient.PostForm(searchURL.String(), data)
    if err != nil {
        return 0, err
    }
    defer resp.Body.Close()

    if resp.StatusCode != http.StatusOK {
        return 0, fmt.Errorf("search start failed: %s", resp.Status)
    }

    var result struct {
        ID int `json:"id"`
    }
    if err := json.NewDecoder(resp.Body).Decode(&result); err != nil {
        return 0, err
    }
    return result.ID, nil
}
```

// GetSearchResults retrieves results for a given search ID.


```

func (c *Client) GetSearchResults(searchID int, limit, offset int) ([]models.Torrent, error) {
    resultsURL := c.BaseURL.ResolveReference(&url.URL{Path: "/api/v2/search/" + strconv.Itoa(searchID)})
    q := resultsURL.Query()
    q.Set("id", strconv.Itoa(searchID))
    if limit > 0 {
        q.Set("limit", strconv.Itoa(limit))
    }
    if offset > 0 {
        q.Set("offset", strconv.Itoa(offset))
    }
    resultsURL.RawQuery = q.Encode()

    resp, err := c.HTTPClient.Get(resultsURL.String())
    if err != nil {
        return nil, 0, err
    }
    defer resp.Body.Close()

    if resp.StatusCode != http.StatusOK {
        return nil, 0, fmt.Errorf("failed to get results: %s", resp.Status)
    }

    var response struct {
        Results []models.Torrent `json:"results"`
        Total    int                  `json:"total"`
        Status   string               `json:"status"`
    }
    if err := json.NewDecoder(resp.Body).Decode(&response); err != nil {
        return nil, 0, err
    }
    return response.Results, response.Total, nil
}

// StopSearch stops a running search.
func (c *Client) StopSearch(searchID int) error {
    stopURL := c.BaseURL.ResolveReference(&url.URL{Path: "/api/v2/search/" + strconv.Itoa(searchID)})
    data := url.Values{}
    data.Set("id", strconv.Itoa(searchID))

    resp, err := c.HTTPClient.PostForm(stopURL.String(), data)
    if err != nil {
        return err
    }
    defer resp.Body.Close()
    if resp.StatusCode != http.StatusOK {
        return fmt.Errorf("failed to stop search: %s", resp.Status)
    }
    return nil
}

// ListPlugins returns all installed search plugins.
func (c *Client) ListPlugins() ([]map[string]interface{}, error) {
    pluginsURL := c.BaseURL.ResolveReference(&url.URL{Path: "/api/v2/search/"})
    resp, err := c.HTTPClient.Get(pluginsURL.String())

```

```

    if err != nil {
        return nil, err
    }
    defer resp.Body.Close()
    if resp.StatusCode != http.StatusOK {
        return nil, fmt.Errorf("failed to list plugins: %s", resp.Status)
    }
    var plugins []map[string]interface{}
    if err := json.NewDecoder(resp.Body).Decode(&plugins); err != nil {
        return nil, err
    }
    return plugins, nil
}

```

5.4 Torrent Management Endpoints

File: internal/client/torrents.go

```
package client
```

```

import (
    "encoding/json"
    "fmt"
    "net/url"

    "github.com/milos85vasic/qBitTorrent-go/internal/models"
)

```

// GetTorrents returns a list of all torrents.

```

func (c *Client) GetTorrents() ([]models.Torrent, error) {
    infoURL := c.BaseURL.ResolveReference(&url.URL{Path: "/api/v2/torrents"})
    resp, err := c.HTTPClient.Get(infoURL.String())
    if err != nil {
        return nil, err
    }
    defer resp.Body.Close()
    if resp.StatusCode != http.StatusOK {
        return nil, fmt.Errorf("failed to get torrents: %s", resp.Status)
    }
    var torrents []models.Torrent
    if err := json.NewDecoder(resp.Body).Decode(&torrents); err != nil {
        return nil, err
    }
    return torrents, nil
}

```

// AddTorrent adds a new torrent from URL or file.

```

func (c *Client) AddTorrent(torrentURL, savepath, category string) error {
    addURL := c.BaseURL.ResolveReference(&url.URL{Path: "/api/v2/torrents/"})
    data := url.Values{}
    data.Set("urls", torrentURL)
    if savepath != "" {
        data.Set("savepath", savepath)
    }
}

```

```

    }
    if category != "" {
        data.Set("category", category)
    }

    resp, err := c.HTTPClient.PostForm(addURL.String(), data)
    if err != nil {
        return err
    }
    defer resp.Body.Close()
    if resp.StatusCode != http.StatusOK {
        return fmt.Errorf("add torrent failed: %s", resp.Status)
    }
    return nil
}

```

6. Core Business Logic Migration

6.1 Merge Search Service

Goal: Replace Python `asyncio.gather` with goroutines to query multiple plugins concurrently and stream results via SSE.

File: `internal/service/merge_search.go`

```

package service

import (
    "context"
    "fmt"
    "sync"
    "time"

    "github.com/milos85vasic/qBitTorrent-go/internal/client"
    "github.com/milos85vasic/qBitTorrent-go/internal/models"
    "github.com/rs/zerolog/log"
)

type MergeSearchService struct {
    qbitClient *client.Client
}

func NewMergeSearchService(qc *client.Client) *MergeSearchService {
    return &MergeSearchService{qbitClient: qc}
}

// SearchResultBatch represents a batch of torrents from a tracker.
type SearchResultBatch struct {
    Tracker string
    Torrents []models.Torrent
    Error error
}

```

```

// Search performs concurrent searches across all specified plugins.
func (s *MergeSearchService) Search(ctx context.Context, req models.SearchReq) (models.SearchResultBatch, error) {
    if len(req.Plugins) == 0 {
        return nil, fmt.Errorf("no plugins specified")
    }

    out := make(chan SearchResultBatch, len(req.Plugins))
    var wg sync.WaitGroup

    for _, plugin := range req.Plugins {
        wg.Add(1)
        go func(p string) {
            defer wg.Done()
            s.searchPlugin(ctx, req, p, out)
        }(plugin)
    }

    go func() {
        wg.Wait()
        close(out)
    }()

    return out, nil
}

func (s *MergeSearchService) searchPlugin(ctx context.Context, req models.SearchReq, plugin string) {
    // Create a timeout context for this plugin search
    pluginCtx, cancel := context.WithTimeout(ctx, time.Duration(s.qbitClient.Timeout))
    defer cancel()

    // Start search on qBittorrent
    searchID, err := s.qbitClient.StartSearch(req.Query, []string{plugin})
    if err != nil {
        out := SearchResultBatch{Tracker: plugin, Error: err}
        return out
    }

    defer func() {
        _ = s.qbitClient.StopSearch(searchID)
    }()

    // Poll for results until completion or context done
    ticker := time.NewTicker(500 * time.Millisecond)
    defer ticker.Stop()

    var allTorrents []models.Torrent
    for {
        select {
        case <- pluginCtx.Done():
            out := SearchResultBatch{Tracker: plugin, Error: pluginCtx.Err()}
            return out
        case <- ticker.C:
            torrents, total, err := s.qbitClient.GetSearchResults(searchID)
            if err != nil {
                out := SearchResultBatch{Tracker: plugin, Error: err}
                return out
            }
            allTorrents = append(allTorrents, torrents...)
            if total == 0 {
                return nil
            }
        }
    }
}

```



```

// Health endpoint
r.GET("/bridge/health", func(c *gin.Context) {
    c.JSON(http.StatusOK, gin.H{"status": "ok"})
})

// Download endpoint
r.GET("/bridge/download", downloadHandler(cfg))

r.Run(":" + cfg.BridgePort)
}

func downloadHandler(cfg *config.Config) gin.HandlerFunc {
    return func(c *gin.Context) {
        tracker := c.Query("tracker")
        torrentURL := c.Query("url")
        if tracker == "" || torrentURL == "" {
            c.JSON(http.StatusBadRequest, gin.H{"error": "tracker and url"}
            return
        }

        // Load tracker-specific auth
        auth, ok := cfg.PrivateTrackerConfig[tracker]
        if !ok {
            c.JSON(http.StatusNotFound, gin.H{"error": "tracker not config"}
            return
        }

        // Create HTTP client with cookie jar
        jar, _ := cookiejar.New(nil)
        client := &http.Client{
            Jar:      jar,
            Timeout: 30 * time.Second,
            CheckRedirect: func(req *http.Request, via []*http.Request) error {
                // Preserve cookies across redirects
                return nil
            },
        }

        // Perform login
        loginData := url.Values{}
        loginData.Set("username", auth.Username)
        loginData.Set("password", auth.Password)
        loginReq, _ := http.NewRequest("POST", auth.LoginURL, strings.NewReader(loginData.Encode()))
        loginReq.Header.Set("Content-Type", "application/x-www-form-urlencoded")
        loginReq.Header.Set("User-Agent", "Mozilla/5.0")

        loginResp, err := client.Do(loginReq)
        if err != nil {
            c.JSON(http.StatusInternalServerError, gin.H{"error": "login failed"})
            return
        }
        defer loginResp.Body.Close()
        if loginResp.StatusCode != http.StatusOK {
            body, _ := io.ReadAll(loginResp.Body)
            c.JSON(loginResp.StatusCode, gin.H{"error": string(body)})
            return
        }
        c.JSON(http.StatusOK, gin.H{"status": "login successful"})
    }
}

```

```

        c.JSON(http.StatusUnauthorized, gin.H{"error": fmt.Sprintf("lo
        return
    }

    // Now download the torrent file
    downloadReq, _ := http.NewRequest("GET", torrentURL, nil)
    downloadReq.Header.Set("User-Agent", "Mozilla/5.0")
    downloadResp, err := client.Do(downloadReq)
    if err != nil {
        c.JSON(http.StatusInternalServerError, gin.H{"error": "downloa
        return
    }
    defer downloadResp.Body.Close()

    if downloadResp.StatusCode != http.StatusOK {
        c.JSON(downloadResp.StatusCode, gin.H{"error": "torrent not fo
        return
    }

    // Stream the torrent file to the client
    c.Header("Content-Disposition", "attachment; filename=torrent.torr
    c.Header("Content-Type", "application/x-bittorrent")
    _, err = io.Copy(c.Writer, downloadResp.Body)
    if err != nil {
        log.Error().Err(err).Msg("failed to stream torrent file")
    }
}
}
}

```

6.3 Serverâ€™Sent Events (SSE) Streaming

File: internal/service/sse_broker.go

```

package service

import (
    "fmt"
    "io"
    "sync"

    "github.com/gin-gonic/gin"
    "github.com/rs/zerolog/log"
)

type SSEBroker struct {
    clients map[chan string]bool
    mu      sync.RWMutex
}

func NewSSEBroker() *SSEBroker {
    return &SSEBroker{
        clients: make(map[chan string]bool),
    }
}

```

```

}

// Subscribe adds a client channel and returns a function to unsubscribe.
func (b *SSEBroker) Subscribe() (chan string, func()) {
    b.mu.Lock()
    defer b.mu.Unlock()
    ch := make(chan string, 10)
    b.clients[ch] = true
    return ch, func() {
        b.mu.Lock()
        defer b.mu.Unlock()
        delete(b.clients, ch)
        close(ch)
    }
}

// Publish sends a message to all subscribed clients.
func (b *SSEBroker) Publish(event string, data string) {
    b.mu.RLock()
    defer b.mu.RUnlock()
    msg := fmt.Sprintf("event: %s\ndata: %s\n\n", event, data)
    for ch := range b.clients {
        select {
        case ch <- msg:
        default:
            // Client is slow, drop message
            log.Warn().Msg("dropping SSE message for slow client")
        }
    }
}

// GinHandler returns a Gin handler that streams SSE.
func (b *SSEBroker) GinHandler() gin.HandlerFunc {
    return func(c *gin.Context) {
        c.Header("Content-Type", "text/event-stream")
        c.Header("Cache-Control", "no-cache")
        c.Header("Connection", "keep-alive")
        c.Header("Access-Control-Allow-Origin", "*")

        ch, unsubscribe := b.Subscribe()
        defer unsubscribe()

        clientGone := c.Request.Context().Done()
        for {
            select {
            case <-clientGone:
                log.Info().Msg("SSE client disconnected")
                return
            case msg := <-ch:
                _, err := io.WriteString(c.Writer, msg)
                if err != nil {
                    return
                }
            }
            c.Writer.Flush()
        }
    }
}

```



```
    }  
  }  
}
```

7. HTTP API Endpoints with Gin

7.1 Main Application Setup

File: cmd/qbittorrent-proxy/main.go

```
package main  
  
import (  
    "context"  
    "encoding/json"  
    "net/http"  
    "time"  
  
    "github.com/gin-gonic/gin"  
    "github.com/milos85vasic/qBitTorrent-go/internal/api"  
    "github.com/milos85vasic/qBitTorrent-go/internal/client"  
    "github.com/milos85vasic/qBitTorrent-go/internal/config"  
    "github.com/milos85vasic/qBitTorrent-go/internal/middleware"  
    "github.com/milos85vasic/qBitTorrent-go/internal/models"  
    "github.com/milos85vasic/qBitTorrent-go/internal/service"  
    "github.com/rs/zerolog/log"  
)  
  
func main() {  
    cfg := config.Load()  
  
    // Initialize qBittorrent client  
    qbitClient, err := client.NewClient(cfg.QBittorrentURL, cfg.QBittorrentUser, cfg.QBittorrentPass)  
    if err != nil {  
        log.Fatal().Err(err).Msg("failed to connect to qBittorrent")  
    }  
  
    // Initialize services  
    searchService := service.NewMergeSearchService(qbitClient)  
    sseBroker := service.NewSSEBroker()  
  
    // Gin router  
    r := gin.Default()  
    r.Use(middleware.CORS())  
    r.Use(middleware.Logger())  
  
    // Health check  
    r.GET("/health", func(c *gin.Context) {  
        c.JSON(http.StatusOK, gin.H{"status": "ok"})  
    })  
}
```

```

// API v1 group
v1 := r.Group("/api/v1")
{
    // Search endpoint with SSE streaming
    v1.POST("/search", func(c *gin.Context) {
        var req models.SearchRequest
        if err := c.ShouldBindJSON(&req); err != nil {
            c.JSON(http.StatusBadRequest, gin.H{"error": err.Error()})
            return
        }

        // Set up SSE
        c.Header("Content-Type", "text/event-stream")
        c.Header("Cache-Control", "no-cache")
        c.Header("Connection", "keep-alive")
        c.Writer.Flush()

        ctx, cancel := context.WithTimeout(c.Request.Context(), time.D
        defer cancel()

        results, err := searchService.Search(ctx, req)
        if err != nil {
            sseBroker.Publish("error", err.Error())
            return
        }

        for batch := range results {
            if batch.Error != nil {
                sseBroker.Publish("error", batch.Error.Error())
                continue
            }
            data, _ := json.Marshal(batch)
            sseBroker.Publish("results", string(data))
            c.Writer.Flush()
        }
        sseBroker.Publish("done", "{}")
    })

    // Hook management endpoints (stub)
    hooks := v1.Group("/hooks")
    {
        hooks.GET("", api.ListHooks)
        hooks.POST("", api.CreateHook)
        hooks.DELETE("/:id", api.DeleteHook)
    }

    // Private tracker bridge proxy (if needed)
    v1.GET("/bridge/download", api.ProxyBridgeDownload(cfg))
}

// Start server
log.Info().Str("port", cfg.ServerPort).Msg("starting server")
if err := r.Run(":" + cfg.ServerPort); err != nil {
    log.Fatal().Err(err).Msg("server failed")
}

```

```
    }  
}
```

7.2 Hook Handlers Example

File: internal/api/hooks.go

```
package api
```

```
import (  
    "net/http"  
  
    "github.com/gin-gonic/gin"  
    "github.com/milos85vasic/qBitTorrent-go/internal/models"  
)
```

```
var hooksStore = make(map[string]models.Hook) // In production, use a database
```

```
func ListHooks(c *gin.Context) {  
    hooks := make([]models.Hook, 0, len(hooksStore))  
    for _, h := range hooksStore {  
        hooks = append(hooks, h)  
    }  
    c.JSON(http.StatusOK, hooks)  
}
```

```
func CreateHook(c *gin.Context) {  
    var hook models.Hook  
    if err := c.ShouldBindJSON(&hook); err != nil {  
        c.JSON(http.StatusBadRequest, gin.H{"error": err.Error()})  
        return  
    }  
    hook.ID = generateID()  
    hook.Enabled = true  
    hooksStore[hook.ID] = hook  
    c.JSON(http.StatusCreated, hook)  
}
```

```
func DeleteHook(c *gin.Context) {  
    id := c.Param("id")  
    if _, ok := hooksStore[id]; !ok {  
        c.JSON(http.StatusNotFound, gin.H{"error": "hook not found"})  
        return  
    }  
    delete(hooksStore, id)  
    c.Status(http.StatusNoContent)  
}
```

8. Middleware & Cross-Cutting Concerns

8.1 CORS Middleware

File: internal/middleware/cors.go

```
package middleware
```

```
import "github.com/gin-gonic/gin"
```

```
func CORS() gin.HandlerFunc {
    return func(c *gin.Context) {
        c.Writer.Header().Set("Access-Control-Allow-Origin", "*")
        c.Writer.Header().Set("Access-Control-Allow-Credentials", "true")
        c.Writer.Header().Set("Access-Control-Allow-Headers", "Content-Type")
        c.Writer.Header().Set("Access-Control-Allow-Methods", "POST, OPTIONS")

        if c.Request.Method == "OPTIONS" {
            c.AbortWithStatus(204)
            return
        }
        c.Next()
    }
}
```

8.2 Structured Logging with Zerolog

File: internal/middleware/logging.go

```
package middleware
```

```
import (
    "time"
```

```
    "github.com/gin-gonic/gin"
    "github.com/rs/zerolog/log"
)
```

```
func Logger() gin.HandlerFunc {
    return func(c *gin.Context) {
        start := time.Now()
        path := c.Request.URL.Path
        raw := c.Request.URL.RawQuery

        c.Next()

        latency := time.Since(start)
        if raw != "" {
            path = path + "?" + raw
        }

        log.Info().
            Str("method", c.Request.Method).
```

```

        Str("path", path).
        Int("status", c.Writer.Status()).
        Dur("latency", latency).
        Str("client_ip", c.ClientIP()).
        Msg("request")
    }
}

```

9. Testing Strategy & Implementation

9.1 Unit Tests for qBittorrent Client

File: internal/client/client_test.go

```

package client

import (
    "net/http"
    "net/http/httptest"
    "testing"
)

func TestLogin(t *testing.T) {
    server := httptest.NewServer(http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {
        if r.URL.Path == "/api/v2/auth/login" && r.Method == "POST" {
            http.SetCookie(w, &http.Cookie{Name: "SID", Value: "test-sid"})
            w.WriteHeader(http.StatusOK)
            return
        }
        w.WriteHeader(http.StatusNotFound)
    }))
    defer server.Close()

    client, err := NewClient(server.URL, "user", "pass")
    if err != nil {
        t.Fatalf("expected no error, got %v", err)
    }
    if !client.IsAuthenticated() {
        t.Error("expected client to be authenticated")
    }
}

```

9.2 Integration Test for Search Flow

File: internal/api/search_test.go

```

package api

import (
    "bytes"
    "encoding/json"
    "net/http"

```

```

    "net/http/httptest"
    "testing"

    "github.com/gin-gonic/gin"
    "github.com/milos85vasic/qBitTorrent-go/internal/models"
)

func TestSearchEndpoint(t *testing.T) {
    gin.SetMode(gin.TestMode)
    r := gin.Default()
    // ... setup dependencies and routes

    body, _ := json.Marshal(models.SearchRequest{
        Query: "linux",
        Plugins: []string{"all"},
    })
    req, _ := http.NewRequest("POST", "/api/v1/search", bytes.NewReader(body))
    req.Header.Set("Content-Type", "application/json")
    w := httptest.NewRecorder()
    r.ServeHTTP(w, req)

    if w.Code != http.StatusOK {
        t.Errorf("expected 200, got %d", w.Code)
    }
    // Additional SSE parsing tests...
}

```

9.3 Run Tests

```

go test ./... -v
go test -race ./...

```

10. Docker & Deployment Configuration

10.1 Multi-Stage Dockerfile

File: Dockerfile

```

# Build stage
FROM golang:1.23-alpine AS builder
WORKDIR /app
COPY go.mod go.sum ./
RUN go mod download
COPY . .
RUN CGO_ENABLED=0 GOOS=linux go build -o /qbittorrent-proxy ./cmd/qbittorrent-proxy
RUN CGO_ENABLED=0 GOOS=linux go build -o /webui-bridge ./cmd/webui-bridge

# Final stage
FROM alpine:3.19
RUN apk --no-cache add ca-certificates tzdata
WORKDIR /app
COPY --from=builder /qbittorrent-proxy .

```

```
COPY --from=builder /webui-bridge .
COPY .env.example .env
EXPOSE 7187 7188
CMD ["/app/qbittorrent-proxy"]
```

10.2 Docker Compose

File: docker-compose.yml

```
version: '3.8'
services:
  qbittorrent:
    image: linuxserver/qbittorrent:latest
    container_name: qbittorrent
    environment:
      - PUID=1000
      - PGID=1000
      - TZ=Europe/London
      - WEBUI_PORT=8080
    volumes:
      - ./config/qbittorrent:/config
      - ./downloads:/downloads
      - ./plugins:/config/qBittorrent/nova3/engines
    ports:
      - "8080:8080"
      - "6881:6881"
      - "6881:6881/udp"
    restart: unless-stopped

  backend:
    build: .
    container_name: qbittorrent-proxy
    ports:
      - "7187:7187"
      - "7188:7188"
    environment:
      - QBITTORRENT_URL=http://qbittorrent:8080
      - QBITTORRENT_USERNAME=admin
      - QBITTORRENT_PASSWORD=adminadmin
    depends_on:
      - qbittorrent
    volumes:
      - ./plugins:/plugins:ro
    restart: unless-stopped

  frontend:
    build: ./frontend
    container_name: qbittorrent-frontend
    ports:
      - "80:80"
    depends_on:
      - backend
    restart: unless-stopped
```

10.3 Update start.sh

The existing start.sh script should be updated to build the Go binaries and Docker images. Example snippet:

```
#!/bin/bash
echo "Building Go backend..."
go build -o bin/qbittorrent-proxy ./cmd/qbittorrent-proxy
go build -o bin/webui-bridge ./cmd/webui-bridge

echo "Building Angular frontend..."
cd frontend && npm install && npm run build --prod && cd ..

echo "Starting services with Docker Compose..."
docker-compose up -d --build
```

11. Plugin System Compatibility Assurance

11.1 How Plugins Are Called

The Go backend **does not execute** the Python Nova3 plugins. It delegates all search tasks to the qBittorrent client via its Web API. The qBittorrent client has its own Python interpreter and runs the plugins exactly as before.

11.2 Verification Steps

1. **List plugins** via Go backend: GET /api/v2/search/plugins (proxied to qBittorrent).
2. **Start a search** with POST /api/v1/search – this internally calls qBittorrent’s /api/v2/search/start.
3. **Observe results** – the SSE stream contains torrents parsed by the plugins.

11.3 Plugin Installation

The install-plugin.sh script remains unchanged. It copies plugin files into the volume mounted to qBittorrent’s nova3/engines directory. The Go backend does not interfere.

11.4 Edge Cases

- **Private trackers requiring login:** The webui-bridge service handles authentication and provides a download endpoint. The Go backend proxies download requests to this bridge.
 - **Plugin timeouts:** The Go service respects the PLUGIN_TIMEOUT environment variable and cancels the context if a plugin hangs.
-

12. Post-Migration Validation Checklist

#	Item	Verification Method
1		Run curl against each endpoint

#	Item	Verification Method
	All <code>/api/v1/*</code> endpoints respond with expected status codes	
2	SSE search stream delivers results from multiple trackers	Use browser EventSource or <code>curl -N</code>
3	Hook registration and triggering works	Register a test webhook and verify payload
4	Private tracker downloads succeed	Test with a configured tracker (e.g., IPT)
5	Angular frontend loads and interacts correctly	Manual UI testing
6	Existing Nova3 plugins are listed and used	Check search results contain expected tracker names
7	Logs are structured and contain correlation IDs	Inspect Docker logs
8	No memory leaks or goroutine leaks under load	Run with <code>pprof</code> and monitor
9	Docker images build successfully	<code>docker-compose build</code>
10	Environment variables are correctly parsed	Add debug endpoint to dump config

13. Appendix: Fileâ€‘byâ€‘File Migration Map

Python File	Go Equivalent	Notes
<code>main.py</code>	<code>cmd/qbittorrent-proxy/main.go</code>	Gin router setup, service wiring
<code>app/api/search.py</code>	<code>internal/api/search.go</code> (and <code>service/merge_search.go</code>)	SSE streaming, concurrent plugin calls
<code>app/api/hooks.py</code>	<code>internal/api/hooks.go</code>	CRUD for webhooks
<code>app/api/bridge.py</code>	<code>cmd/webui-bridge/main.go</code>	Private tracker auth & download
<code>app/models/*.py</code>	<code>internal/models/*.go</code>	Structs with JSON tags
<code>app/core/config.py</code>	<code>internal/config/config.go</code>	

Python File	Go Equivalent	Notes
		Environment loading with godotenv
webui-bridge.py	cmd/webui-bridge/main.go	Standalone microservice
requirements.txt	go.mod	Dependency management
openapi.json	N/A (source of truth)	Used for API contract validation
start.sh	scripts/build.sh + updated start.sh	Build and orchestrate

End of Migration Guide

This document provides a complete, actionable plan. Follow each section sequentially, and refer to the file map and code examples to ensure every line of Python logic is translated into Go. The result will be a performant, maintainable backend that works seamlessly with your existing Angular frontend and qBittorrent plugins.